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1 #include <stdio.h>
2 struct priority_scheduling {
3     char process_name;
4     int burst_time;
5     int waiting_time;
6     int turn_around_time;
7     int priority;
8 };
9 int main() {
10     int number_of_process;
11     int total = 0;
12     int position;
13     float average_waiting_time;
14     float average_turnaround_time;
15     printf("Enter the total number of Processes: ");
16     scanf("%d", &number_of_process);
17     struct priority_scheduling process[number_of_process];
18     struct priority_scheduling temp_process;
19     int ASCII_number = 65;
20     printf("\nPlease Enter the Burst Time and Priority of each process:\n");
21     for (int i = 0; i < number_of_process; i++) {
22         process[i].process_name = (char)ASCII_number;
23         printf("\nEnter details of process %c\n", process[i].process_name);
24         printf("Enter the burst time: ");
25         scanf("%d", &process[i].burst_time);
26         printf("Enter the priority: ");
27         scanf("%d", &process[i].priority);
28         ASCII_number++;
29     }
30     for (int i = 0; i < number_of_process; i++) {
31         position = i;
32         for (int j = i + 1; j < number_of_process; j++) {
33             if (process[j].priority > process[position].priority) {
34                 position = j;
35             }
36         }
37         temp_process = process[i];
38         process[i] = process[position];
39         process[position] = temp_process;
40     }
41     process[0].waiting_time = 0;
42     for (int i = 1; i < number_of_process; i++) {
43         process[i].waiting_time = 0;
44         for (int j = 0; j < i; j++) {
45             process[i].waiting_time += process[j].burst_time;
46         }
47         total += process[i].waiting_time;
48     }
49     average_waiting_time = (float)total / number_of_process;
50     total = 0;
51     printf("\n\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
52     for (int i = 0; i < number_of_process; i++) {
53         process[i].turn_around_time =
54             process[i].burst_time + process[i].waiting_time;
55         total += process[i].turn_around_time;
56         printf("%c\t%d\t\t\t%d\t\t\t%d\n",
57             process[i].process_name,
58             process[i].burst_time,
59             process[i].waiting_time,
60             process[i].turn_around_time);
61     }
62     average_turnaround_time = (float)total / number_of_process;
63     printf("\nAverage Waiting Time : %f", average_waiting_time);
64     printf("\nAverage Turnaround Time : %f\n", average_turnaround_time);
65     return 0;
66 }

```

Enter the total number of Processes: 3

Please Enter the Burst Time and Priority of each process:

Enter details of process A

Enter the burst time: 2

Enter the priority: 1

Enter details of process B

Enter the burst time: 10

Enter the priority: 3

Enter details of process C

Enter the burst time: 2

Enter the priority: 6

Process	Burst Time	Waiting Time	Turnaround Time
C	2	0	2
B	10	2	12
A	2	12	14

Average Waiting Time : 4.666667

Average Turnaround Time : 9.333333

=== Code Execution Successful ===